

Proposed Solution to #5823 SSMJ

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Statement of the Problem: Prove that if $0 < a \leq b$, then:

$$\left(\int_a^b e^{-x^2} dx \right)^2 \geq \left(\int_{\frac{a+b}{2}}^b e^{-x^2} dx + \int_{\sqrt{ab}}^b e^{-x^2} dx \right) \left(\int_a^{\frac{a+b}{2}} e^{-x^2} dx + \int_a^{\sqrt{ab}} e^{-x^2} dx \right)$$

Solution of the Problem:

Let $F(x)$ be the real antiderivative of e^{-x^2} . Then the given problem can be restated as:

$$[F(b) - F(a)]^2 \geq \left[F(b) - F\left(\frac{a+b}{2}\right) + F(b) - F(\sqrt{ab}) \right] \left[F\left(\frac{a+b}{2}\right) - F(a) + F(\sqrt{ab}) - F(a) \right]$$

$$[F(b) - F(a)]^2 \geq \left[2F(b) - F\left(\frac{a+b}{2}\right) - F(\sqrt{ab}) \right] \left[F\left(\frac{a+b}{2}\right) + F(\sqrt{ab}) - 2F(a) \right]$$

$$\begin{aligned} F^2(b) - 2F(b)F(a) + F^2(a) &\geq 2F(b) \left[F\left(\frac{a+b}{2}\right) + F(\sqrt{ab}) \right] - 4F(b)F(a) - \left[F\left(\frac{a+b}{2}\right) + F(\sqrt{ab}) \right]^2 \\ &\quad + 2F(a) \left[F\left(\frac{a+b}{2}\right) + F(\sqrt{ab}) \right] \end{aligned}$$

which can be rewritten as

$$\begin{aligned} [F(b) + F(a)]^2 + \left[F\left(\frac{a+b}{2}\right) + F(\sqrt{ab}) \right]^2 &\geq 2 \left[F\left(\frac{a+b}{2}\right) + F(\sqrt{ab}) \right] [F(b) + F(a)] \\ \left[F(b) + F(a) - F\left(\frac{a+b}{2}\right) - F(\sqrt{ab}) \right]^2 &\geq 0 \end{aligned}$$

Since square of a real number cannot be negative, the given statement is true which proves the original statement we began with, hence concluding the proof.